



# Web Deployment & Management

NT208 - Web Programming



# How to deploy a website?

You need to have:

1. A domain name
2. A hosting (VPS/Cloud/etc...)
3. Website's source code (your final project)
4. A brain



# 1. Domain

- Free domain (limited features)
- Paid domain (full access)

# Find cheap domain names for your website

Hosting, Email and SSL solutions are available to set up your online presence all under one roof.

SEARCH DOMAIN

**.com** \$13.95

**.biz** \$5.75 ~~\$15.55~~

 **.BEAUTY** \$1.99 ~~\$11.99~~

**.shop** \$2.99 ~~\$26.99~~

 **.link** \$3.99 ~~\$8.59~~

**.eu** \$2.70 ~~\$9.99~~

Compare:

**.COM**

 **NameSilo**  
Consider. Cheap, easy and secure

**\$13.95**

Go Daddy

**\$15.99**

Google Domains

**\$17.00**

Name.com

**\$12.99**



## REGISTER

[Domain Name Search](#)[Bulk Search](#)[Pricing](#)

## DOMAIN ACTIONS

[Transfer](#)[Renew](#)

## AFTERMARKET

[Premium](#)[Expiring](#)[All](#)

## FEATURED DOMAINS

**.PRO**~~\$24.99~~ / **\$3.99****.IO**~~\$49.99~~ / **\$34.99****.COM**~~\$13.99~~ / **\$11.99**Search nearly 300 unique  
domain extensions

FIND YOUR DOMAIN

Search domain names



- Paid domains are not so expensive



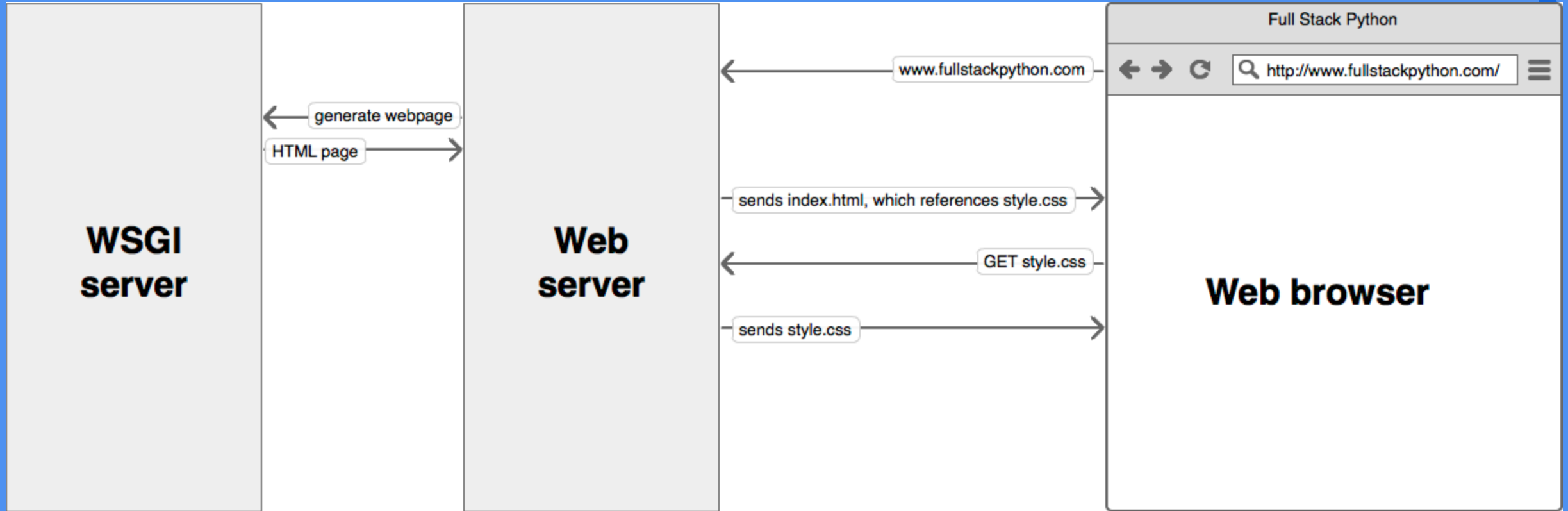
# 1. Hosting

- Free hosting (limited features)
- Paid hosting (higher privileges)
- Cloud Service
- Dedicated Server (full access)

| Cloud VPS Plans | RAM  | CPU | SSD Storage | Transfer | Price    | figure                      |
|-----------------|------|-----|-------------|----------|----------|-----------------------------|
| cVPS-1          | 1GB  | 1   | 25GB        | 1TB      | \$5/mo   | <a href="#">Get Started</a> |
| cVPS-2          | 2GB  | 1   | 50GB        | 2TB      | \$10/mo  | <a href="#">Get Started</a> |
| cVPS-3          | 3GB  | 1   | 60GB        | 2TB      | \$15/mo  | <a href="#">Get Started</a> |
| cVPS-4          | 4GB  | 2   | 80GB        | 4TB      | \$20/mo  | <a href="#">Get Started</a> |
| cVPS-6          | 6GB  | 3   | 120GB       | 4TB      | \$30/mo  | <a href="#">Get Started</a> |
| cVPS-8          | 8GB  | 4   | 160GB       | 5TB      | \$50/mo  | <a href="#">Get Started</a> |
| cVPS-16         | 16GB | 6   | 320GB       | 6TB      | \$80/mo  | <a href="#">Get Started</a> |
| cVPS-32         | 32GB | 8   | 640GB       | 7TB      | \$160/mo | <a href="#">Get Started</a> |

| Features/Hosts                                            | Hostinger                       | InMotion Hosting             | A2 Hosting           | DreamHost                              | Bluehost             |
|-----------------------------------------------------------|---------------------------------|------------------------------|----------------------|----------------------------------------|----------------------|
| Starting Pricing (Monthly)                                | ~\$6 - \$22                     | ~\$15 - \$75                 | ~\$40 - \$100        | ~\$10 - \$80                           | ~\$30 - \$70         |
| Renewal Pricing (Monthly)                                 | ~\$8 - \$50                     | ~\$45 - \$198                | ~\$65 - \$204        | ~\$10 - \$80                           | ~\$40 - \$120        |
| CPU Cores Range                                           | 1 - 8                           | 4 - 32                       | 2 - 8                | N/A                                    | 2 - 4                |
| Allowable RAM (GB)                                        | 4 - 32                          | 4 - 32                       | 4 - 32               | 1 - 8                                  | 2 - 8                |
| Unlimited Bandwidth                                       | No; 1 TB to 8 TB                | Yes, except for VPS 4 GB RAM | No; 2 TB to 4 TB     | Yes                                    | Yes                  |
| Unlimited Storage Space                                   | No; 50 GB to 400 GB             | No; 90 GB to 540 GB          | No; 150 GB to 450 GB | No; 30 GB to 240 GB                    | No; 120 GB to 240 GB |
| Dedicated IP                                              | 1                               | 2 - 10                       | 2                    | 1                                      | No                   |
| Uptime                                                    | 99.997%                         | 99.995%                      | 99.993%              | 99.992%                                | 99.963%              |
| 24/7 Customer Support (Live Chat, Phone, & Email/Tickets) | 24/7 via Live Chat & Email Only | Yes                          | Yes                  | 24/7 via Live Chat, Email, & Call Back | Yes                  |





okluntuwsl@OKLIWIN: ~

```
okluntuwsl@OKLIWIN:~$ echo "hello !" > hello.txt  
okluntuwsl@OKLIWIN:~$ vim hello.txt
```

```
$ sudo apt update
```

Copy

With the update complete, you can install Git:

```
$ sudo apt install git
```

Copy

You can confirm that you have installed Git correctly by running the following command and checking that you receive relevant output.

```
$ git --version
```

Copy

Output

```
git version 2.25.1
```

# Important commands

- apt update
- apt upgrade
- apt dist-upgrade

# Essential commands

- `apt install -y curl xclip git webp`
- `apt install -y build-essential libssl-dev`
- `apt install -y tcl8.5`
- `apt install -y software-properties-common python-software-properties`
- `apt install -y apt-transport-https`
- `apt install -y software-properties-common dirmngr`

# Essential commands

- `apt install -y build-essential`
- `apt install -y libncurses5-dev libncursesw5-dev libreadline6-dev`
- `apt install -y libdb-dev libgdbm-dev libsqlite3-dev libssl-dev`
- `apt install -y libbz2-dev libexpat1-dev liblzma-dev zlib1g-dev`
- `apt install -y libcurl4-openssl-dev libpam0g-dev libsystemd-dev`
- `apt install -y zlib1g-dev libsqlite3-dev tk-dev`
- `apt install -y libssl-dev openssl`
- `apt install -y libffi-dev`

# Essential commands

- `apt install -y python-certbot-nginx`
- `apt install -y libxml2-dev libxslt1-dev libxslt-dev`
- `apt install -y python-dev python-setuptools python3-dev python3-setuptools`
- `apt install -y python-imaging`
- `apt install -y python-pip`
- `pip install virtualenv`

# Essential commands

- `apt install -y fail2ban`
- `apt install -y iptables-persistent`
- `apt install -y memcached libmemcached-dev gettext silversearcher-ag git zsh`
- `apt install -y nethogs iftop`



# Essential commands

```
pyenv activate myenv  
pip install -U pip wheel  
pip install django  
  
# cài các gói phụ thuộc nếu cần  
pip install -r requirements.txt
```

Create the App's name:

```
django-admin startproject myproject  
cd myproject  
python manage.py migrate  
python manage.py createsuperuser  
python manage.py runserver
```

# Setup WSGI server (Back-end)

- pip install gunicorn
- sudo apt install supervisor
- vim supervisor.ini

# Setup WSGI server (Back-end)

```
[unix_http_server]
file=/home/myuser/configs/supervisor.sock ; (the path to the socket file)

[supervisord]
logfile=/home/myuser/logs/supervisord.log ; (main log file;default $CWD/supervisord.log)
logfile_maxbytes=50MB ; (max main logfile bytes b4 rotation;default 50MB)
logfile_backups=10 ; (num of main logfile rotation backups;default 10)
loglevel=error ; (log level;default info; others: debug,warn,trace)
pidfile=/home/myuser/configs/supervisord.pid ; (supervisord pidfile;default supervisord.pid)
nodaemon=false ; (start in foreground if true;default false)
minfds=32000 ; (min. avail startup file descriptors;default 1024)
minprocs=32000 ; (min. avail process descriptors;default 200)

[rpcinterface:supervisor]
supervisor.rpcinterface_factory = supervisor.rpcinterface:make_main_rpcinterface

[supervisorctl]
serverurl=unix:///home/myuser/configs/supervisor.sock

[program:django]
command=/home/myuser/venv/bin/gunicorn myproject.wsgi:application -b 0.0.0.0:8800
directory=/home/myuser/django/myproject
priority=1
autostart=true
```

# Setup Webserver (Nginx)

- `sudo apt install nginx`
- `sudo systemctl status nginx`
- `vim nginx.conf`

# Setup Webserver (Nginx)

```
upstream django {  
    server 127.0.0.1:8800;  
}  
  
server {  
    listen 80;  
    server_name django-install.vietdev.com;  
  
    access_log off;  
    error_log off;  
  
    gzip_disable "msie6";  
    gzip_vary on;  
    gzip_proxied any;  
    gzip_comp_level 6;  
    gzip_buffers 16 8k;  
    gzip_http_version 1.1;  
    gzip_min_length 256;  
    gzip_types text/plain text/css application/json application/javascript application/x-javascript text/xml appl:  
    gzip on;  
  
    root /home/myuser/django;  
    index index.html index.htm;  
  
    location / {  
        proxy_next_upstream error timeout invalid_header http_500 http_502 http_503 http_504;  
        proxy_redirect off;  
        proxy_buffering off;  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
        proxy_http_version 1.1;  
        proxy_read_timeout 60s;  
        proxy_send_timeout 180s;  
        proxy_pass http://django;  
    }  
}
```

# 3. Domain name & DNS



Sales: +65 3158 3954    Support ▾     ▾

[Solutions](#)   [Products](#)   [Pricing](#)   [Resources](#)   [Partners](#)   [Why Cloudflare](#)

   [Sign up](#)   [Under attack?](#)   [Log in](#)

## Pricing

[Our plans & pricing](#) >

[Enterprise plan](#)

[Compare all plans](#)

[Need help choosing?](#)

### Zero Trust & SASE

ZTNA, CASB, SWG, RBI, email security, & more

### Application Security & Performance

DDoS, WAF, CDN, DNS, load balancing, & more

### Network Security & Performance

WAN, FWaaS and DDoS protection

### Developer Platform

Build & deploy serverless code

×

### 3. Domain name

## NameServer Manager



You can use the form below to set nameservers for the domains listed above. The first 2 nameservers are required, and you can optionally add up to 11 additional secondary nameservers for a total of 13 possible nameservers.

The nameservers you want to use should have been provided by your web hosting company or ISP, and are typically formatted like "NS1.HOST.COM".

Please do not enter the IP addresses for the name servers.



Please note that entering in third-party nameservers below will prevent domain forwarding and domain parking from working.

#### Name Server Entry

##### Default NameServers

You will need to enter our default nameservers to utilize such services as parking, forwarding and DNS management.  
[Click here to select our default nameservers](#)

[Click here to select our Premium nameservers](#)

Name Server 1:  \*

Name Server 2:  \*

[Add additional name servers](#)

SUBMIT

# 3. Domain name & DNS

| Type                      | Name (required)                                     | IPv4 address (required)        | Proxy status                                                     | TTL             |
|---------------------------|-----------------------------------------------------|--------------------------------|------------------------------------------------------------------|-----------------|
| <div>A</div> <div>▼</div> | <div>django-install</div> <div>Use @ for root</div> | <div>1.1.1.1</div> <div></div> | <div><div><div>✓</div><div></div></div><div> Proxied</div></div> | <div>Auto</div> |



django

View [release notes](#) for Django 4.2



The install worked successfully! Congratulations!

You are seeing this page because [DEBUG=True](#) is in your settings file and you have not configured any URLs.



**Django Documentation**

Topics, references, & how-to's



**Tutorial: A Polling App**

Get started with Django



**Django Community**

Connect, get help, or contribute



## 4. Management

- Google Search Console
- Google Analytics
- Similarweb
- Ahrefs
- Google Pagespeed insights
- ChatGPT
- Etc..

e

example.com

▼

- 🏠

Overview
- 🔍

URL inspection

- Performance
- G

Search results
- ✳️

Discover
- 📰

Google News

- Index
- 📄

Coverage
- 🗺️

Sitemaps
- 🗑️

Removals

- Enhancements
- 📏

Core Web Vitals
- 📱

Mobile Usability
- 📁

Breadcrumbs
- 📁

Events
- 📁

Logos
- 📁

Products

Performance on Discover

📄

Date: Last 28 days

✎

+

NEW

Last updated: 4 hours ago ?

⬇️

EXPORT

☒ Total clicks

☐ Total impressions

☐ Avg. CTR

5.5K

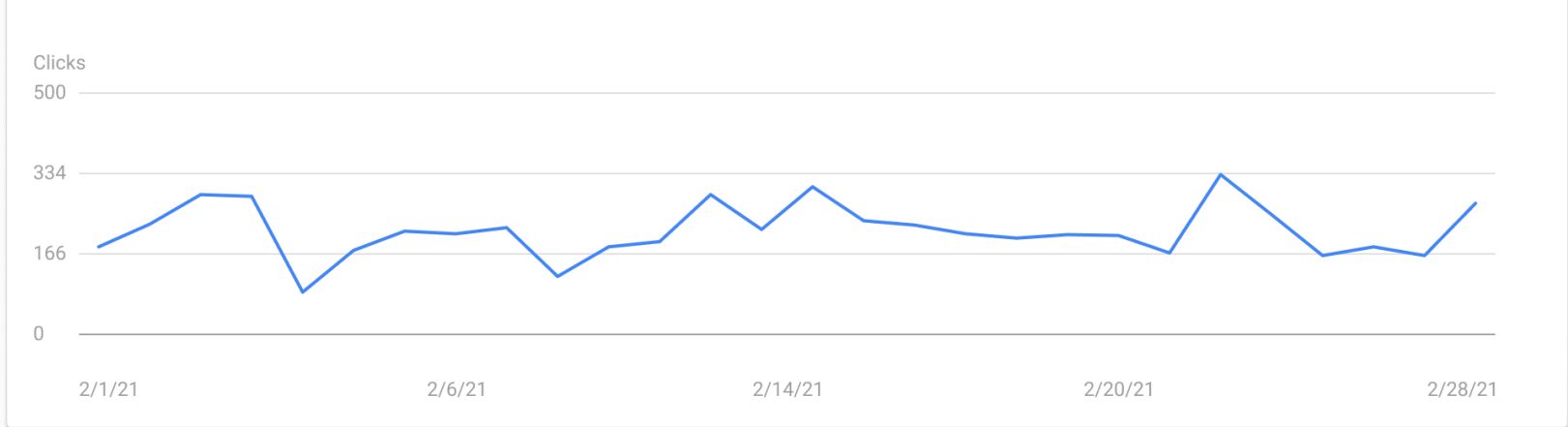
120K

4.6%

?

?

?



| PAGES                             | COUNTRIES | DISCOVER APPEARANCE | DATES |                      |
|-----------------------------------|-----------|---------------------|-------|----------------------|
| Page                              |           |                     |       | ↓ Clicks Impressions |
| https://www.example.com/article-2 |           |                     |       | 1,257 18,763         |
| https://www.example.com/article-1 |           |                     |       | 1,070 19,147         |
| https://www.example.com/article-5 |           |                     |       | 987 12,745           |

https://morningscor... ▼

- Overview
- URL inspection

Performance

- Search results
- Discover

Indexing

- Pages
- Video pages
- Sitemaps
- Removals

Experience

- Page experience
- Core web vitals
- Mobile Usability

Enhancements

- Breadcrumbs
- FAQ
- How-to

## Page experience overview

Core web vitals ?

101

Failing URLs

Mobile Usability ?

1

Failing URLs

HTTPS ?

Good

Your site uses HTTPS.

## Desktop

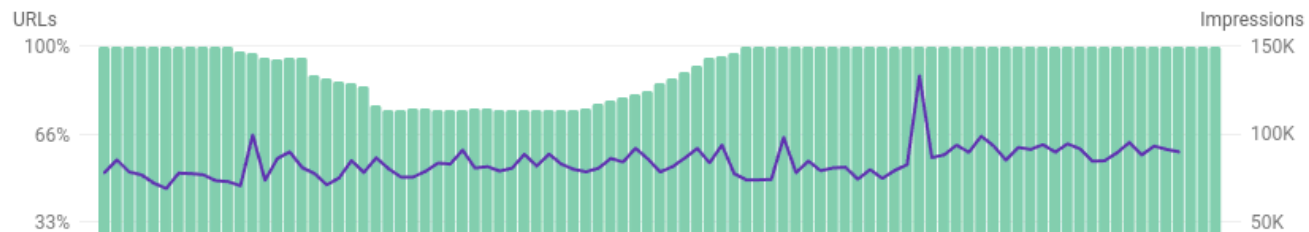
Good URLs ?

100%

Total impressions of good URLs ?

7.38M

About chart



Overview Overview: dandomain.dk/ DanDomain | Webshop – Domæne – Web – Hosting

- Overview
- Overview 2.0
- Site structure
- Calendar

- Backlink profile
- Backlinks
- Broken backlinks
- Referring domains
- Anchor
- Internal backlinks
- Referring IPs
- Link intersect

- Organic search
- Organic keywords
- Top pages
- Top subfolders
- Top subdomains
- Organic competitors
- Competing domains
- Competing pages
- Content gap

- Paid search
- Paid keywords
- Ad

Monthly volume All countries Changes: Last year

**Backlink profile**  


DR<sup>i</sup>  
**85**  
AR 4,919 ▼214



UR<sup>i</sup>  
**43**

Backlinks<sup>i</sup>  
**1.2M** +237K  
All time 11.3M

Ref. domains<sup>i</sup>  
**27.4K** +6.1K  
All time 154K

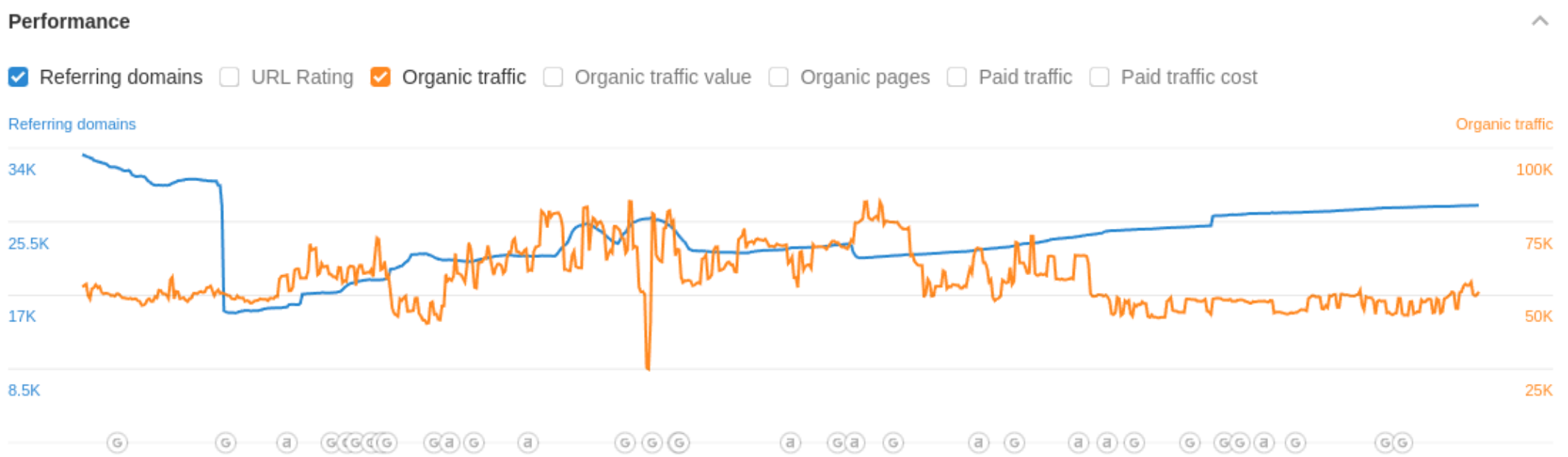
**Organic search**  
Keywords<sup>i</sup>  
**5.2K** -394  
Top 3 287 -54

Traffic<sup>i</sup>  
**51.3K** -15.7K  
Value \$73.5K -25.2K

**Paid search**  
Keywords<sup>i</sup>  
**32** -72  
Ads 35 -56

Traffic<sup>i</sup>  
**5.1K** -5.4K  
Cost \$4.6K -9.6K

Metrics Competitors Countries Years 1M 6M 1Y 2Y All Daily



70

uit.edu.vn

PERFORMANCE

15/30

SEO

30/30

MOBILE

20/30

SECURITY

5/10

Is your website slowing you down?

Create and manage beautiful website pages that get traffic and convert leads with HubSpot CMS Free.

Get the Free CMS

No credit card needed

PAGE SIZE

3.2MB

A respectable pace. Well played.

The heavier the site page, the slower the load. For optimal performance, try to keep page size below 3MB.

PAGE REQUESTS

42

Serious room for improvement.

The more HTTP requests your website makes, the slower it becomes. Try reducing the number of files your site loads.

PAGE SPEED

6.1SEC

Might try speeding things up a bit.

Best-in-class webpages should become interactive within 5.3 seconds. Any slower and visitors will abandon your site, reducing conversions and sales.

BROWSER CACHING

PASS

Wowee. Your web caching is world class.

Browser caching speeds up your website by storing frequently used content in local memory.

MINIMAL PAGE REDIRECTS

PASS

Straight to the point.

Multiple redirects can make your site load slower. Aim for no more than one redirect.

IMAGE SIZE

FAIL

Out of place.

Images can take a long time to load. Use responsive images or SVGs to optimize your images for different screen sizes.

9/3/20XX

Presentation Title

30



Mobile



Desktop



📅 Latest 28-day collection period

🕒 Full visit durations

📱 Various mobile devices

🌐 Various network connections

👥 Many samples ([Chrome UX Report](#))

🔄 All Chrome versions



## Diagnose performance issues



Performance



Accessibility



Best Practices



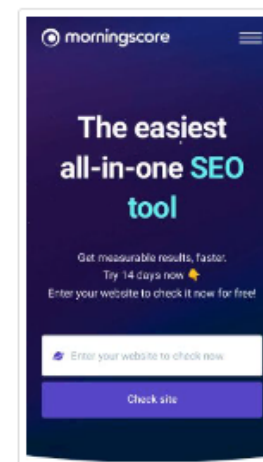
SEO



## Performance

Values are estimated and may vary. The [performance score is calculated](#) directly from these metrics. [See calculator.](#)

▲ 0–49    ■ 50–89    ● 90–100



# Bài tập

- Nhóm bạn được giao nhiệm vụ xây dựng một trang web rút gọn link (URL shortener), tương tự như bit.ly hay tinyurl.com.
- Hãy lựa chọn một gói VPS (Virtual Private Server) phù hợp cho dự án này trong giai đoạn khởi đầu. Dựa vào bảng giá hoặc thông tin của các nhà cung cấp.
- **Yêu cầu:**
  - Nêu rõ thông số gói VPS bạn chọn:  
(RAM, CPU, SSD, băng thông, giá tiền,...)
  - Giải thích lý do bạn chọn gói đó:  
(Khối lượng truy cập ước tính? Nhu cầu tài nguyên? Mức chi phí chấp nhận được?)
  - Nêu ít nhất 2 gói VPS KHÔNG được chọn và giải thích lý do loại bỏ.  
(Quá yếu? Quá đắt? Không cần thiết cho giai đoạn đầu?...)





Thank you